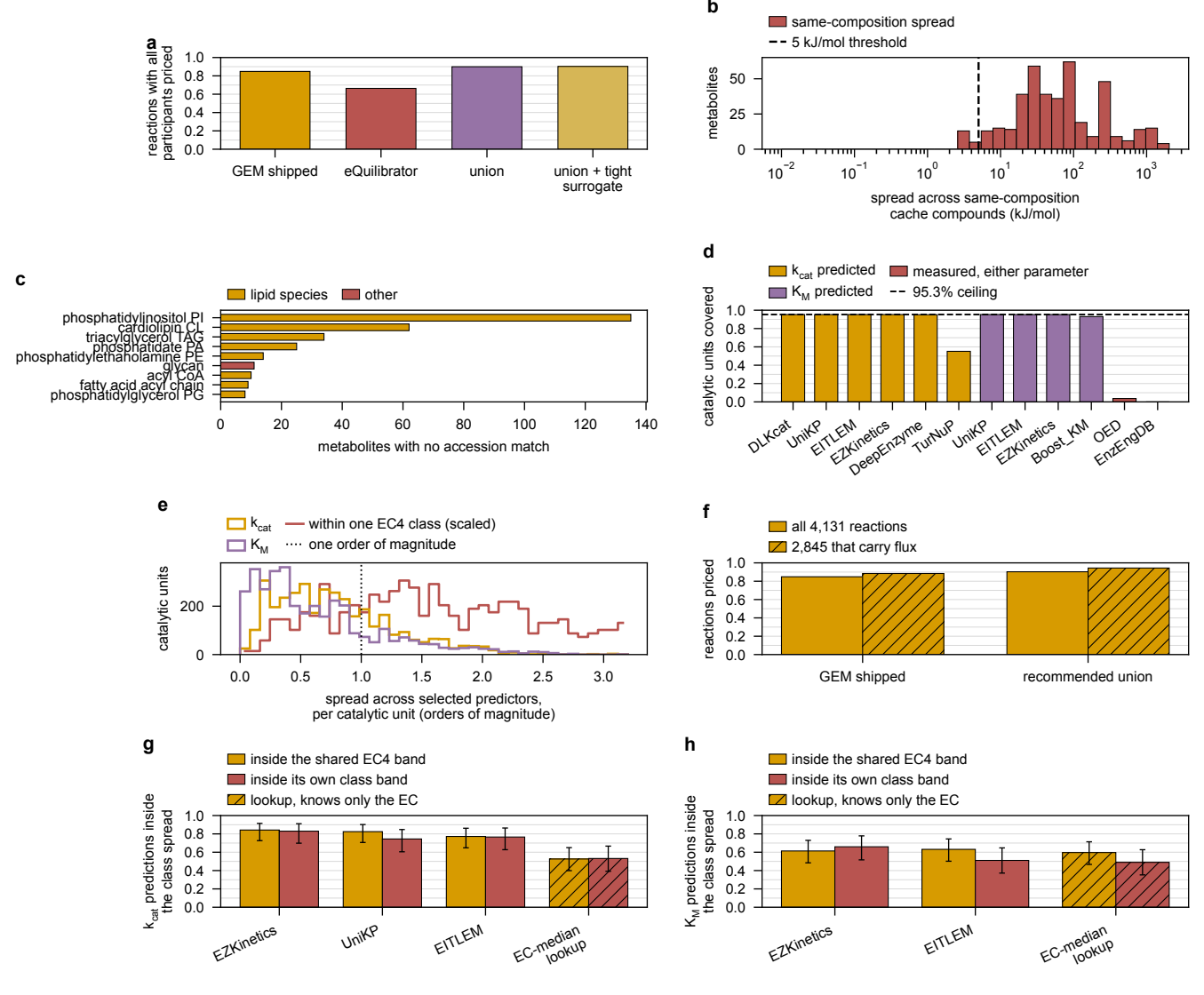
a
reactions with all participants priced
1.0
0.8
0.6
0.4
0.2
0.0
GEM shipped
eQuilibrator
union
union + tight surrogate
b
same-composition spread
5 kJ/mol threshold
metabolites
50
25
0
$10^{-2}$
$10^{-1}$
$10^{0}$
$10^{1}$
$10^{2}$
$10^{3}$
spread across same-composition cache compounds (kJ/mol)
c
lipid species
other
phosphatidylinositol PI
cardiolipin CL
triacylglycerol TAG
phosphatidate PA
phosphatidylethanolamine PE
glycan
acyl CoA
fatty acid acyl chain
phosphatidylglycerol PG
0
20
40
60
80
100
120
140
metabolites with no accession match
d
$k_{cat}$ predicted
measured, either parameter
$K_M$ predicted
95.3% ceiling
catalytic units covered
1.0
0.8
0.6
0.4
0.2
0.0
DLKcat
UniKP
EITLEM
EZKinetics
DeepEnzyme
TurNuP
UniKP
EITLEM
EZKinetics
Boost_KM
OED
EnzEngDB
e
$k_{cat}$
$K_M$
within one EC4 class (scaled)
one order of magnitude
catalytic units
200
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
spread across selected predictors, per catalytic unit (orders of magnitude)
f
all 4,131 reactions
2,845 that carry flux
reactions priced
1.0
0.8
0.6
0.4
0.2
0.0
GEM shipped
recommended union
g
inside the shared EC4 band
inside its own class band
lookup, knows only the EC
$k_{cat}$ predictions inside the class spread
1.0
0.8
0.6
0.4
0.2
0.0
EZKinetics
UniKP
EITLEM
EC-median lookup
h
inside the shared EC4 band
inside its own class band
lookup, knows only the EC
$K_M$ predictions inside the class spread
1.0
0.8
0.6
0.4
0.2
0.0
EZKinetics
EITLEM
EC-median lookup